

WORKSHOP BASICS

Tools & Workshop Skills

Measure carefully. Join cleanly. Leave the bench safer than you found it.

THE BENCH ROUTINE

Every connection has a check.

1. **Measure** before touching the circuit.
2. **Join** the parts with heat and solder.
3. **Protect** the finished connection.



Use the multimeter without guessing

https://youtu.be/ts0EVc9vXcs?si=VAHfErAzgUeZid_C

Choose the function and the jack before the probes touch the circuit.

BENCH ROUTINE

1

Choose

Voltage, current, resistance, or continuity.

3

Read

Touch carefully and watch the display.

2

Connect

Black probe in **COM**; red probe in **V Ω** or **A/mA**.

4

Reset

Remove power before changing connections.



Warning: Current uses the **A/mA** jack and must be measured in series — never like voltage.

Soldering joins the component to the PCB

<https://youtu.be/QKbJxytERvg?si=LembyOcntE5HEtza>

Heat the pad and lead together, then let the solder flow into the joint.

Touch the iron to the pad and component lead.

Feed solder into the heated joint.

Pull solder away first.

Lift the iron and let the joint cool.

GOOD JOINT



STOP AND FIX

Avoid: bridges, blobs, dull cracked joints, and lifted pads.

Inspect before power

Desoldering removes a connection cleanly

https://youtu.be/N_dvf45hN6Y?si=BZJJLEWw_NGQHmCD

Free a component or correct a soldering mistake without damaging the pad or board.

PUMP

- 1 **Heat** the joint.
- 2 **Position** the nozzle.
- 3 **Press** the release and pull molten solder away.



WICK

- 1 **Place** braid on the joint.
- 2 **Heat** through the copper braid.
- 3 **Lift** the braid after it absorbs solder.



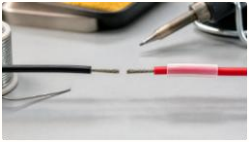
Basic rule: use brief heat, keep the tip clean, and stop if the pad starts to lift.

Heat shrink protects exposed wire

<https://youtu.be/VgnHuJGocZI?si=fWHzw7Yr-jafq3qn>

Slide it on before the joint, then center and heat it evenly afterward.

BEFORE



Slide the tubing onto one wire. Keep it away from the heat.

AFTER



Center and heat evenly. The sleeve grips the insulation.

CHECK



No burnt tubing.



No exposed conductor.



No movement at the joint.

FINAL BENCH CHECK

Workshop safety is part of every connection



Protect

Wear eye protection. Keep loose clothing, hair, and cables away from the iron.



Park

Rest the hot iron in its stand. Never leave it tip-down or unattended.



Ventilate

Work with ventilation, wash hands after soldering, and keep food and drink away.



Power off

Disconnect power before changing probe leads, desoldering, or moving a circuit.



Check the meter

Confirm the range and jack. Never measure current like voltage.



Reset the bench

Iron parked, power off, scraps cleared, tools returned.

Angle grinder

Cut, grind, and clean metal edges

PURPOSE

Purpose: Remove excess metal, cut stock, and smooth welds.

HOW TO USE

Basic operation: Fit the correct disc, hold the grinder with two hands, start it clear of the work, then guide it steadily through the cut or across the surface.



SAFETY

Safety: Wear eye and face protection, hearing protection, gloves, and closed footwear. Check the guard and disc for damage. Keep sparks away from people, cables, and flammables.



Drill machine

Make accurate holes safely

PURPOSE

Purpose: Drill holes for fasteners, brackets, and cable routes.

HOW TO USE

Basic operation: Mark and center-punch the hole, clamp the work securely, select the bit and speed, then apply steady pressure without forcing the tool.



SAFETY

Safety: Wear eye protection and tie back long hair or loose clothing. Remove the chuck key before starting. Never hold small work by hand, and wait for the bit to stop before clearing swarf.

